National Aeronautics and Space Administration

A Researcher's Guide to:

INTERNATIONAL SPACE STATION

Plant Science

The Lab is Open

Soaring 250 miles above Earth, the International Space Station (ISS) is a modern wonder of the world, combining the efforts of 15 countries and thousands of scientists, engineers, and technicians. The ISS is a magnificent platform and laboratory for all kinds of research to improve life on Earth, enable future space exploration, and understand the universe. This guide is intended to help potential researchers plan and carry out plant experiments aboard the ISS, provide an overview of plant growth chambers available for use, and discuss the integrated spaceflight environment. This includes using the continuous freefall of the microgravity environment to study the role of gravity and other spaceflight environment effects on plant growth and metabolism in low Earth Orbit.

Cosmonaut Gennady Padalka harvests radishes from the Lada Plant Chamber (June 22, 2009).

Unique Features of the ISS Research Environment

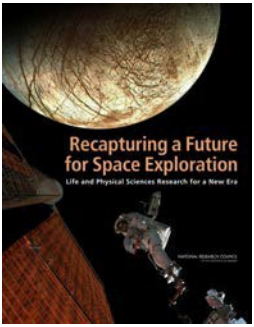
1. **Microgravity**, or weightlessness, alters many observable phenomena within the physical and life sciences. Systems and processes affected by microgravity include surface wetting and interfacial tension, multiphase flow and heat transfer, multiphase system dynamics, solidification, and fire phenomena and combustion. Microgravity induces a vast array of changes in organisms ranging from bacteria to humans, including global alterations in gene expression and 3-D aggregation of cells into tissue-like architecture.

2. **Extreme conditions** on the ISS space environment include exposure to extreme heat and cold cycling, ultra-vacuum, atomic oxygen, high energy radiation, and island biology isolation. Testing and qualification of materials exposed to these extreme conditions have provided data to enable the manufacturing of long-life reliable components used on Earth as well as in the world's most sophisticated satellite and spacecraft components

3. **Low-Earth orbit** at 51 degrees inclination and a 90-minute orbit affords ISS a unique vantage point with an altitude of approximately 240 miles (400 kilometers) and an orbital path over 90 percent of the Earth's population. This path can provide improved spatial resolution and variable lighting conditions compared to the sun-synchronous orbits of typical Earth remote-sensing satellites.

Why Use the ISS as a Laboratory?

The ISS provides a unique platform from which to understand how plants respond to gravity and the spaceflight environment. This knowledge is important because plants and their microbial communities are critical components of bioregenerative life support systems (BLSS) that will enable humans to thrive in space.



The National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a NewEra," (http://www.nap.edu/catalog/13048.html) recognized the significance of understanding plant responses in microgravity for both space exploration and terrestrial applications. A new report schedule for release in the summer of 2023 will have amended recommendations.

The 2011 report had a number of recommendations for plant research, including the following:

1. The establishment of a robust spaceflight program of research analyzing plant and microbial growth and physiological responses to multiple stimuli encountered in the spaceflight environments.

2. The establishment of a research program aimed at demonstrating the roles of microbial-plant systems in long-term life support systems.

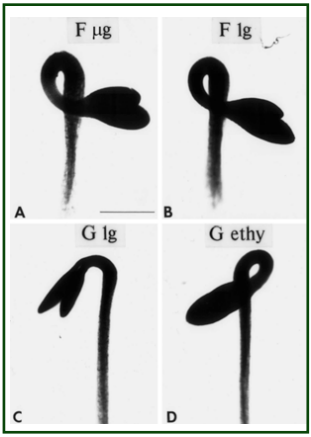
A number of knowledge gaps exist that are well-suited to be addressed using the current capabilities of spaceflight hardware and analytical capabilities on the ISS. These gaps point to the necessity of maximizing the science return from each spaceflight opportunity and the need for both basic and applied research as well as transcriptomic, proteomic, and metabolomic analysis in model plants. These studies as well as life cycle research are necessary for developing effective horticultural techniques and validating environmental conditions to enable the incorporation of plants in BLSS. A partial listing of these research questions is provided in Table 1.

LESSONS LEARNED FROM PLANT RESEARCH ON ISS

INSIGHTS INTO FUNDAMENTAL PLANT BIOLOGICAL PROCESSES

Interactions Between Gravity and Other Space Environmental Stimuli

Gravity has profound effects on the growth and development of plants. Most notable is the gravity-induced downward growth of roots and the upward growth of shoots. This phenomenon, called gravitropism, has been the subject of scientific investigation since the work of Charles Darwin (Darwin, 1881). Gravitropism is of broad interest to plant scientists because it is a target trait in breeding programs that aim to improve crop nutrient and water uptake from the soil, as well as enhance light capture for photosynthesis. (Chin and Blancaflor, 2022). However, the constant presence of gravity on Earth limits the breadth of questions that can be asked about the phenomenon, particularly those that pertain to its interaction with other environmental stimuli and plant movements.



Light micrographs of hypocotyls of Arabidopsis seedlings from the F µg (A), ground controls (B = F 1g; C = G 1g), and ethylene control (D = G ethy). Note the anomalous hook that developed in hypocotyls of the light seedlings (both F µg and F 1g) and in the ethylene ground control. A typical hypocotyl hook, characteristic of dark- grown seedlings, developed in G 1g seedlings. Bar = 500 µm. (Source: NASA)

Biosatellite II experiments in the 1960s were among the first to document the effects of microgravity on plant movements. A common observation in these experiments was the occurrence of epinasty, a process in which petioles and leaves bend down (Johnson and Tibbitts, 1968). Early studies also revealed a high incidence of epinasty in plants grown on clinostats (Brown et al., 1974). Epinasty is regulated for the most part by ethylene, and levels of this gaseous plant hormone have been reported to be elevated in the Space Shuttle cabin (Guisinger and Kiss, 1999; Campbell et al., 2001) as well as in the Russian Mir Space Station. Results from the Space Shuttle and ISS have revealed microgravity-induced changes in seedling morphology and reduced expression of genes encoding

ethylene biosynthetic enzymes (Kiss et al., 2000; Wakabayashi et al., 2017). These observations support the hypothesis that the epinasty that occurs during spaceflight is in part due to ethylene.

Shannon Lucid inspects wheat grown in Svet plant growth chamber on the Mir space station. (Source: NASA)

More recent studies on the ISS have led to additional insights into other plant movements that are typically dampened because of the ubiquitous nature of gravity on Earth. For example, circumnutation, a type of oscillatory plant movement, was shown to persist in weightless conditions with sunflower seedlings (*Helianthus annuus*), proving it was a true endogenous rhythm, just as Darwin suggested. But it is also dependent on gravity (Kobayashi et al., 2019). Work with coleoptile circumnutation of rice (*Oryza. Sativa*) gravitropism mutants validated earlier observations that microgravity damps circumnutation of inflorescence stems in the model plant *Arabidopsis thaliana* (Johnsson et al., 2009). By contrast, skewing and waving, phenomena observed predominantly in roots of *Arabidopsis thaliana,* or thale cress, growing on hard substrates, appear to be less influenced by gravity. This conclusion is supported by the observation that *A. thaliana* roots in the Space Shuttle and ISS still skew (Millar et al., 2011; Paul et al., 2012; Nakashima et al., 2014). Analysis of *A. thaliana* mutants in space revealed that the cytoskeleton is a component of pathways that modulate root skewing in microgravity (Nakashima et al., 2014; Califar et al., 2020).

Research on hydrotropism and phototropism has also advanced from work on the ISS. Cucumber (*Cucumis sativus*) roots displayed a stronger hydrotropic response on the ISS than on Earth. The hydrotropic response is dependent on the redistribution of the plant hormone auxin (Morohashi et al., 2017). Regarding phototropism, work with *A. thaliana* seedlings on the ISS uncovered modifications in the directional growth of roots in response to blue and red light, phenomena typically masked by gravity on Earth (Millar et al., 2010; Vandenbrink et al., 2016). Interaction between red light phototropism and fractional gravity on root growth was demonstrated by Valbuena et al. (2018). Taken together, the microgravity environment of the ISS has provided unique opportunities to expand the breadth of research on gravitropism and other tropistic responses that could benefit space crop production, as well as agriculture on Earth.